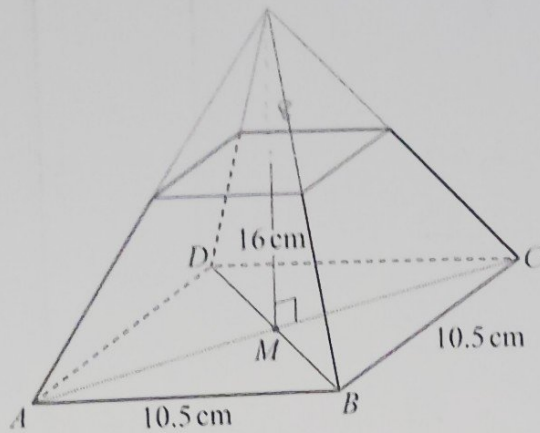


(b)



NOT TO SCALE

The diagram shows a frustum of the pyramid $OABCD$.
The height of the frustum is 16 cm.

Calculate the volume of the frustum.

Volume of large pyramid

$$= 882 - 78.5$$

$$= 808.5 \text{ cm}^3$$

$$= \frac{1}{3} Ah$$

$$= \frac{1}{3} \times 110.25 \times 24$$

$$= 882 \text{ cm}^3$$

$$LSF = 2$$

$$ASF = 2^2 = 4$$

$$\dots\dots\dots 808.5 \dots\dots\dots \text{cm}^3 [5]$$

Volume of small pyramid

$$= \frac{1}{3} \times 27.6 \times 8$$

$$= 73.5 \text{ cm}^3$$

Question 21 is on page 16.



21 $\frac{16^{5m}}{4} = 64^{2n}$

Find m in terms of n .

$$\frac{16^{5m}}{4}$$

$$16^{5m} = 256^{2n}$$

$$16^{5m} = (16^2)^{2n}$$

$$16^{5m} = 16^{4n}$$

$$5m = 4n$$

$$m = \frac{4n}{5}$$

$$m = \frac{4n}{5} \quad [3]$$

22 $I = \frac{V}{R}$

V is 50, correct to the nearest 10.

R is 13, correct to the nearest integer.

Calculate the upper bound of I .

$$V \begin{matrix} 0 \\ 50 \end{matrix} \quad 45 \quad 50 \quad 55$$

$$R \begin{matrix} 0 \\ 13 \end{matrix} \quad 12 \quad 13 \quad 14$$

$$I = \frac{55}{12}$$

$$= 4.6$$

$$\dots\dots\dots 4.6 \dots\dots\dots [3]$$

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